



DELHI PUBLIC SCHOOL NEWTOWN
SESSION 2024-25
MONDAY TEST

CLASS : X
SUBJECT: MATHEMATICS

FULL MARKS: 40
DATE: 30.10.24

General Instructions:

- The paper consists of three printed pages.
- All questions are compulsory.
- Copy the question number carefully before answering the questions.

SECTION: A

1. [1 × 8 = 8]
- i) Given $a = 3\sec^2\theta$ and $b = 3\tan^2\theta - 2$. The value of $(a - b)$ is:
a) 1 b) 2 c) 3 d) 5
- ii) Which of the following lines cut the positive x-axis and positive y-axis at equal distances from the origin?
a) $3x + 3y = 6$ b) $5x + 10y = 10$ c) $-x + y = 1$ d) $10x + 5y = 5$
- iii) If the probability of hitting the target is 0.3, then the probability of not hitting the target is:
a) 0.3 b) 0.5 c) 0.7 d) 1
- iv) At a certain time of the day, the ratio of the height of the pole to the length of its shadow is $1:\sqrt{3}$, then the angle of the sun at that time of the day is:
a) 30° b) 45° c) 60° d) 90°
- v) Assertion (A) : A die is thrown once and the probability of getting an odd number is $\frac{7}{8}$.
Reason (R) : The sample space for odd numbers on a die is $\{1, 3, 5\}$
- (a) Both A and R are true, and R is the correct reason for A.
(b) Both A and R are true, and R is incorrect reason for A.
(c) A is true, R is false.
(d) A is false, R is true.
- vi) If the centre of the circle is (3,5) and the end points of the diameter are (4,7) and (2,y) then the value of y is:
a) -3 b) 3 c) 7 d) 4
- vii) Statement 1: $\sin^2 A + \cos^2 A = 1$ Statement 2: $\sec^2 A + \tan^2 A = 1$
Which of the following is valid?
a) only 1 b) only 2 c) both 1 and 2 d) neither 1 nor 2

viii) Points A(x,y), B(3,-2) and C(4,-5) are collinear. The value of y in terms of x is:

a) $3x - 11$

b) $11 - 3x$

c) $3x - 7$

d) $7 - 3x$

SECTION: B

2.

(a) Factorize: $\sin^3 \theta + \cos^3 \theta$, hence prove the following identity:

$$\frac{\sin^3 \theta + \cos^3 \theta}{\sin \theta + \cos \theta} + \sin \theta \cos \theta = 1$$

(b) Find:

(i) $(\sin \theta + \operatorname{cosec} \theta)^2$

(ii) $(\cos \theta + \sec \theta)^2$

Using the above results prove the following trigonometry identity.

$$(\sin \theta + \operatorname{cosec} \theta)^2 + (\cos \theta + \sec \theta)^2 = 7 + \tan^2 \theta + \cot^2 \theta \quad [4 + 4 = 8]$$

3.

(a) An integer is chosen between 0 and 100. What is the probability that it is

(i) divisible by 7?

(ii) not divisible by 7?

(iii) a multiple of both 2 and 5?

(b) Each of the letter of the word “ARBITRATION” is written on cards and put in a bag. If a card is drawn at random from the bag after shuffling, what is the probability that the letter on the card is:

(i) a consonant

(ii) one of the letters of the word SPRING.

(iii) not a letter from the word TEAR.

(iv) neither a vowel nor a consonant. [3 + 4 = 7]

4. Given the equations of two straight lines, L1 and L2 are $x - y = 1$ and $x + y = 5$ respectively. If L1 and L2 intersects at point Q (3, 2). Find :

(i) the equation of line L3 which is parallel to L1 and has y-intercept 3.

(ii) the value of k, if the line L3 meets the line L2 at a point P (k,4).

(iii) the coordinate of R and the ratio PQ: QR, if line L2 meets X-axis at point R. [4]

5. Two poles AB and PQ are standing opposite each other on either side of a road 200 m wide. From a point R between them on the road, the angles of elevation of the top of the poles AB and PQ are 45° and 40° respectively. If height of AB = 80 m, find the height of PQ correct to the nearest metre. [4]

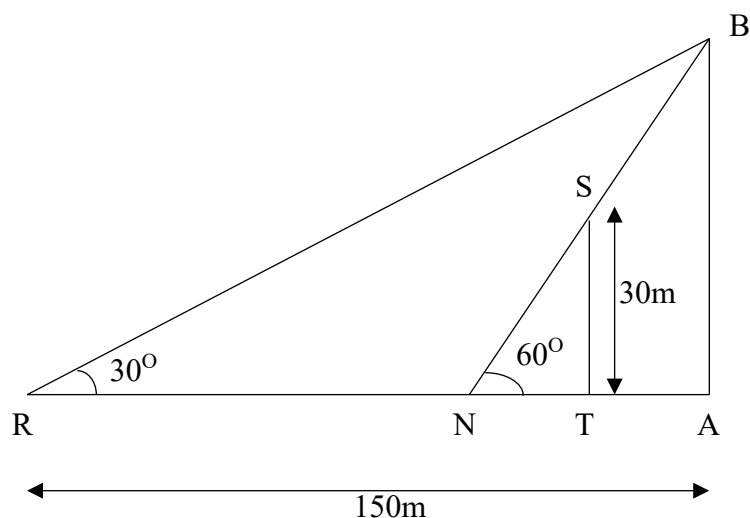
6. A (1, -5), B (2, 2) and C (-2, 4) are the vertices of triangle ABC. Find the equation of :

(i) the median of the triangle through A.

(ii) the altitude of the triangle through B.

(iii) the line through C and parallel to AB. [4]

7.



A tree (TS) of height 30 m stands in front of a tall building (AB). Two friends Rohan and Nisha are standing at R and N respectively, along the same straight line joining the tree and the building (as shown in the diagram). Rohan, standing at a distance of 150 m from the foot of the building, observes the angle of elevation of the top of the building as 30° . Nisha from her position observes that the top of the building and the tree has the same elevation of 60° .

Find the:

- (a) height of the building
- (b) distance between:
 - (i) Nisha and the foot of the building
 - (ii) Rohan and Nisha
 - (iii) Nisha and the tree
 - (iv) building and the tree.

[5]